System Requirements Specification (SRS) for *Avoid Collision during Overtake*

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Table of Contents

Table of Contents **1**
1. Introduction **2**
1.1 Purpose 2
1.2 Document Conventions 2
1.3 Glossary 3
1.4 Intended Audience and Reading Suggestions 4
1.5 Product Scope 4
1.6 References 4
2. Overall Description **5**
2.1 Product Perspective 5
2.2 Product Functions 5
2.3 User Classes and Characteristics 5
2.4 Operating Environment 5
2.5 Design and Implementation Constraints 6
2.6 Assumptions of Use 6
The Operational Scenario 6

Vehicles and Hardware..... 7
The System and Communication.....7
3. System Use Cases.....8
3.1 Collision Avoidance (CA).....8
3.2 Overtake (O)..... 10
3.3 Manage Speed (MS)..... 11
4. System requirements definition.....11
4.1. System of Systems (SoS) Requirements..... 12
4.2. Constituent Systems (CSs) Requirements..... 12
General Requirements for Vehicles B and C..... 12
Specific Operational Logic for Vehicle B..... 12
Specific Operational Logic for Vehicle C.....13
4.3. Interface Requirements..... 13
4.3.1. Relied Upon Interface (RUI) – General Interaction..... 13
4.3.3. Relied Upon Physical Interface (RUPI) – Physical and Environmental Interface.. 13
5. Traceability matrix.....14

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1. Introduction

1.1 Purpose

This document specifies the software and system requirements for Avoid Collisions, an Advanced Driver Assistance System (ADAS) module whose primary goal is to prevent collisions that can occur when a faster vehicle (B) overtakes a slower vehicle (A) on a two-lane undivided road while a vehicle (C) approaches in the opposite direction. The SRS describes a collision avoidance system where the collision avoidance part is performed by ADAS processors installed in cars to let them drive autonomously. It also covers functional and non-functional requirements, use cases and traceability needed to design, implement, verify and validate the system.

1.2 Document Conventions

- *Requirement IDs use the form REQ-<View>-<number> Type, e.g., REQ-FUNC-01 M. Priority levels: High, Medium, Low. Safety-critical requirements are marked [SAFETY].*
- *We use RFC2119 "Key words for use in RFCs to Indicate Requirement Levels".*

1.3 Glossary

- ***Overtaking Maneuver:** A sequence of discrete states of Vehicle B, used by the system to contextualize the risk level.*

1. ***State 0: IN_LANE_FOLLOWING:*** *Vehicle B is in its own lane, following Vehicle A.*
2. ***State 1: LANE_DEPARTURE:*** *The central axis of Vehicle B crosses the centerline. This marks the beginning of the critical phase of the maneuver.*
3. ***State 2: OVERTAKING:*** *Vehicle B entirely occupies the opposing lane.*
4. ***State 3: LANE_REENTRY:*** *Vehicle B begins the maneuver to re-enter its own lane.*
5. ***State 4: MANEUVER_COMPLETE:*** *Vehicle B has completely re-entered its own lane at a safe distance from Vehicle A. The critical phase is concluded.*

- ***Time To Collision (TTC):*** *A temporal metric that estimates the time remaining before a collision between two or more objects, assuming they maintain their current velocities and trajectories. It is calculated as the ratio between the relative distance and the relative velocity between the vehicles.*

$$TTC = \frac{\text{Distance}_{\text{B,C}}}{\text{Speed}_{\text{B}} + \text{Speed}_{\text{C}}}$$

- ***TTC Critical Threshold:*** *The TTC value below which the time remaining until impact is less than the minimum time required for the vehicle to perform a safe complete stop. Falling below this threshold makes the activation of the AEB system mandatory.*
- ***Emergent Behavior:*** *Properties and behaviors observable at the SoS (System of Systems) level (e.g., the near-simultaneous stop in REQ-SoS-02) that are not explicitly designed or controlled by a single CS (Constituent System), but which result from the dynamic and uncoordinated interaction of the CSs.*
- ***Autonomous Emergency Braking (AEB):*** *An active safety function that autonomously triggers the vehicle's maximum braking capacity in response to the determination of an "Inevitable Collision" state.*
- ***Inevitable Collision:*** *A logical state of the system, declared when the TTC is below the critical threshold and, simultaneously, the planning algorithms do not identify any safe evasive trajectory (e.g., acceleration, deceleration, or steering) available for the vehicle.*

1.4 Intended Audience and Reading Suggestions

We can have different type of readers of this document and depending on the case we can give the following suggestions:

- ***System architects / developers:*** *read the entire document, focus on Section 4 (requirements).*
- ***Test engineers:*** *read Sections 3 for use cases coverage.*
- ***Project managers / stakeholders:*** *read Section 1 (overview), Section 2 (product description), and traceability matrix.*
- ***Safety engineers / certification bodies:*** *pay special attention to safety requirements (Sections 4).*

1.5 Product Scope

"Avoid Collisions" is an automotive ADAS component that:

- Monitors relative positions and velocities of vehicles in the same lane and oncoming lanes.
- Detects overtaking maneuvers by rear vehicle (B).
- Determines if an overtake can be completed safely given oncoming traffic (vehicle C).
- Intervenes (active interventions: speed reduction, steering assist or overtaking abort) to avoid collisions during overtakes.

The product is intended to be integrated into the vehicle's ADAS stack (sensors, CAN bus / vehicle network, braking and steering actuators). It is not responsible for vehicle navigation outside overtaking scenarios.

1.6 References

None

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2. Overall Description

2.1 Product Perspective

The system is an ADAS subsystem that receives inputs from:

- *Forward- and rear-facing radar / lidar / cameras (object detection and tracking).*
- *Vehicle odometry (speed, acceleration).*
- *Steering angle sensors and yaw rate sensors.*
- *Vehicle network (CAN/CAN-FD) for brake/engine control commands and to request actuation.*
- *It outputs: actuation requests (brake torque, torque steering assist, engine torque reduction) and logs events for later analysis.*

A high-level block diagram (conceptual): Sensors → Perception → Overtake Decision & Risk Assessment → Intervention Manager → Vehicle Actuators.

2.2 Product Functions

- *F1 - Continuous detection/tracking of nearby vehicles (A, B, C) and lane geometry.*
- *F2 - Relative motion prediction (trajectories, time-to-collision) for vehicles involved.*
- *F3 - Overtake eligibility assessment based on clearance, oncoming traffic, road geometry, legal constraints.*

- *F4 - Active intervention to avoid collision: slow down B, steer assist to remain in lane, or emergency braking.*

2.3 User Classes and Characteristics

- *Vehicle Integrator / OEM Engineer: integrates ADAS module with vehicle.*
- *Safety Engineer / Test Engineer: responsible for verification and validation.*

2.4 Operating Environment

- Two-way undivided roads with two undivided lanes (one lane per direction).
- Ambient temperatures and lighting per automotive standards (-40 °C to +85 °C typical for electronics).
- Road conditions: dry/wet/icy (capabilities degrade with reduced sensor performance).
- Networked automotive electrical system (12 V / 48 V), with required communication buses (CAN, Ethernet optional).

2.5 Design and Implementation Constraints

- *Real-time constraints: perception-to-decision latency must meet safety bounds (see non-functional).*
- *Actuator interface constraints: actuators support level of intervention (e.g., can torque steer or request braking).*
- *Regulatory constraints: must comply with applicable automotive safety standards (ISO 26262), Italian traffic laws for overtaking, the road with 2 undivided lanes has to be considered according to the Italian regulations, the 3 cars have to respect the European regulation for dimensions and emissions.*
- *Sensor limitations: performance must be specified for radar/lidar/cameras under different conditions.*
- *Fail-safe: system must not create hazardous behavior when degraded; must default to warning-only if actuation unavailable.*

2.6 Assumptions of Use

*This chapter defines the fundamental boundary conditions for the project. The system's reliability and safety are guaranteed **only if** the following assumptions hold true. If any of these conditions are not met, the system is not designed to handle the scenario.*

The Operational Scenario

We define the "operating environment" in which the system is tested and validated.

- ***Road Geometry:** The system operates exclusively on **straight, undivided roads** with a single lane per direction of travel (total 2 lanes). Curves, intersections, or multi-lane roads are not managed.*
- ***Actors Involved:** The scenario involves **exactly three vehicles** with predefined roles:*
  - ***Vehicle A:** On the right lane, in front.*
  - ***Vehicle B:** On the right lane, behind Vehicle A.*
  - ***Vehicle C:** On the opposite lane, traveling in the opposite direction.*
- ***Behavior (Speed):** All vehicles travel at a **constant speed**. Specifically: A (50 km/h), B (70 km/h), and C (50 km/h). No sudden accelerations or decelerations are anticipated (other than those commanded by the system itself).*
- ***Maneuver Trigger:** Vehicle B initiates the overtaking maneuver only when it is **sufficiently close** to detect Vehicle A (i.e., A is within its sensor range).*
- ***Feasibility Constraint:** The starting position of Vehicle C is defined such that the initial Time-to-Collision (TTC) provides a theoretical window of opportunity for a safe overtake.*

Vehicles and Hardware

We assume that the onboard hardware is functional and adequate.

- ***Reaction Capability:** The braking systems (particularly of B and C) are **adequate and performant** for the specified speeds, allowing for a safe stop if the maneuver is aborted.*
- ***Perception (Sensors):** The vehicles are equipped with reliable perception systems (radar, LiDAR, cameras) that function correctly and have the necessary range (long-range) to detect other vehicles in a timely manner.*
- ***Third-Party Components (COTS):** It is assumed that any third-party software or hardware component (libraries, sensors, actuators like AEB) **conforms to its specifications** and does not contain hidden vulnerabilities or faults.*
- ***Processing:** We consider the cars characterized by ADAS (Advanced Driver Assistance Systems) processors; the ECU can be considered as outside of the system (actors of the Use Case Diagram).*
- ***Human Factor:** We do not consider human presence inside the vehicles.*

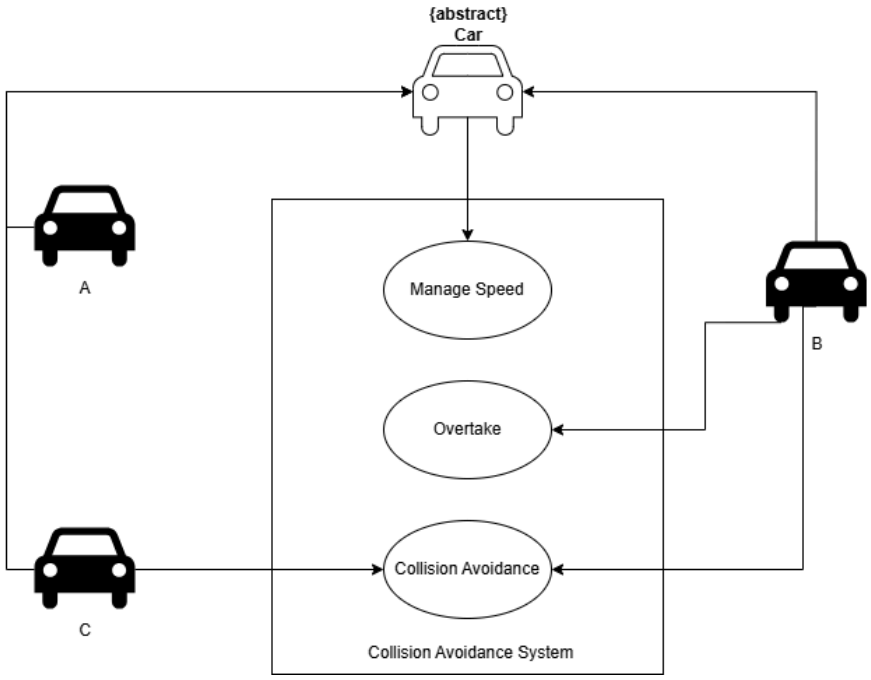
The System and Communication

We define how the subsystems interact and manage problems.

- ***Synchronization:*** *A reliable* ***local time base*** *is available.*
- ***Real-Time:*** *The entire "perception -> decision -> actuation" cycle must adhere to* ***strict hard real-time constraints****. The decision must be made before it is physically too late.*
- ***Clean Communication:*** *All system components communicate only through* ***defined interfaces (APIs)****. There are no "covert channels" or undocumented interactions that could create dangerous emergent behaviors.*
- ***Fault Model (Simple):*** *The system is designed to tolerate* ***benign faults*** *(e.g., a component stops responding or "crashes"). It is* ***not*** *designed to handle Byzantine faults (i.e., components that "go rogue" and send intentionally erroneous or conflicting data).*

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3. System Use Cases



3.1 Collision Avoidance (CA)

Objective - The ECU in the autonomous system of a car solicits the car to slow down until it stops, with the objective to avoid a collision. The braking event occurs only when the car detects another object, that could be the car in the front or the car in the other lane.

Priority – High

Actors - ECU's B and C

Flow of Events

1. **Basic Flow**

   1.1. B and C keep detecting if there are objects in front of them within a certain distance.
   1.2. B doesn't detect anything, so it keeps going.

   1.3. C doesn't detect anything, so it keeps going.

   1.4 Go back to step 1.1

2. **Alternative Flow(1)** - at step 1.2 B detects something

   2.1 The object is going in the same direction (A).

   2.2 B starts the Overtake procedure (O).

   2.3 Go back to step 1.2.

3. **Alternative Flow(2)** - at step 1.2 B detects something

   3.1 The object is coming from a different direction in the same lane B is in (car C).

   3.2 The approaching object detects B too.

   3.3 C and B perform an emergency brake to avoid collision.

4. **Alternative Flow(3)** - at step 1.3 C detects something

   4.1 The object (B) detects C too.

   4.2 C and B perform an emergency brake to avoid collision.

5. **Exception Flow(1)** - malfunction in sensors or actuators in any of the cars

   5.1 The malfunctioning car enters safe mode by slowing down gradually until it stops executing (MS).

Includes - Overtake (O), Manage Speed (MS).

Preconditions - A, B and C need to be far enough at the start to avoid immediate collisions. The use case has to be activated at the start of the system.

Post conditions - Any collision is avoided and B and C can both be still going or staying still depending on the completion or not of the overtake.

Notes/Issues - B, C and A are actually referred to each respective ECU.

3.2 Overtake (O)

Objective - The ECU in the autonomous system of a car solicits the car to slow down until it stops, with the objective to avoid a collision. The braking event occurs only when the car detects another object, that could be the car in the front or the car in the other lane.

Priority – Medium

Actors - ECU B

Flow of Events

1. **Basic Flow**

   1.1. B performs (CA) to check if the opposing lane is empty.

   1.2. B switches lanes while keeping CA active.

   1.3. When B has overtaken A it steers back into its lane

2. **Alternative Flow(1)** - at step 1.1 B receives a signal of “no empty lane” from (CA) before switching lane

   2.1 stops the execution of the overtake procedure and remains in the initial lane

   Alternative Flow(2) - at step 1.2 B receives “collision” signal from (CA) while in the opposite lane

   3.1 Emergency braking is started by (CA).

   3.2 Overtake procedure is dismissed.

   3.3 (C and B perform an emergency brake to avoid collision.)??

3. **Exception Flow(1)**

Includes - Collision Avoidance(CA), Manage Speed (MS).

Preconditions - A and B need to be close enough (10 m) to be able to start the procedure.

Post conditions - B may have been able to overtake A if it didn’t encounter C coming from the other lane while overtaking, in that last case then B and C remain still after the emergency brake. If at the start of the procedure B gets a “no empty lane” signal B aborts the procedure immediately while not even changing lanes, then B just keeps going behind A.

Notes/Issues - None.

3.3 Manage Speed (MS)

Objective – Permits all the ECUs in the car to regulate the speed depending on the situation and permitting the 3 cars to go at a constant speed in a standard situation, i.e. **IN_LANE_FOLLOWING**.

Priority – Medium

Actors – ECU A, B and C

Flow of Events : Loop managing speed.

Includes - None

Preconditions – None.

Post conditions – None.

Notes/Issues - None.

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4. System requirements definition

4.1. System of Systems (SoS) Requirements

The following requirements describe the observable characteristics of the emergent behavior that the system must exhibit.

- **REQ-SoS-01 M:** The System of Systems (SoS) must be the operational scenario where two Constituent Systems (CSs) enter a potential collision trajectory, which triggers their respective individual emergency procedures.
- **REQ-SoS-02 M:** The resolution of a frontal trajectory conflict must manifest as a complete stop of the involved CSs, within the limits imposed by the physical laws of braking.
- **REQ-SoS-03 M:** The stopped state of the vehicles must be the result of independent and non-coordinated computational decisions, made by each CS based on data acquired exclusively from its own sensor apparatus.

4.2. Constituent Systems (CSs) Requirements

General Requirements for Vehicles B and C

- **REQ-CS-GEN-01 M:** Each vehicle must be equipped with a multi-spectral sensor system for forward-facing perception, including long-range (e.g., radar, LiDAR) and short-range (e.g., cameras) sensors.

- **REQ-CS-GEN-02 M:** Each vehicle must have an onboard processing unit capable of executing real-time algorithms for trajectory prediction and "Time To Collision" (TTC) calculation.
- **REQ-CS-GEN-03 M:** Each vehicle must be equipped with an Autonomous Emergency Braking (AEB) system that activates autonomously if the TTC falls below a predefined critical threshold.

Specific Operational Logic for Vehicle B

- **REQ-CS-B-01 M:** Before initiating an overtaking maneuver, the vehicle's system must evaluate, via its own sensors, that the clear space in the opposite lane is sufficient to begin the maneuver safely.
- **REQ-CS-B-02 M:** If, after starting the maneuver, the sensors detect an approaching Vehicle C, the system must calculate the TTC.
- **REQ-CS-B-03 M:** Vehicle B must perform an overtaking maneuver if it identifies A in front of it, according to REQ-CS-B-01 M and REQ-CS-B-02 M.

Specific Operational Logic for Vehicle C

- **REQ-CS-C-01 M:** The vehicle's system must constantly monitor its own lane of travel to detect the presence of obstacles.
- **REQ-CS-C-02 M:** If sensors detect a frontal obstacle (Vehicle B) on its travel path, the onboard algorithm must calculate the TTC.
- **REQ-CS-C-03 M:** If the algorithm determines that a collision is inevitable, it must activate the emergency systems defined in REQ-CS-GEN-03.

4.3. Interface Requirements

4.3.1. Relied Upon Interface (RUI) – General Interaction

Defines the requirements for the interface between the ADAS subsystem (the product specified in this SRS) and the rest of the vehicle architecture in B and C (Sensors and Actuators).

- **REQ-RUI-01 M:** The ADAS subsystem interface (RUI) must be fully operational (receiving data from sensors and ready to send commands to actuators) as soon as Vehicle B enters State 1 (LANE_DEPARTURE).
- **REQ-RUI-02 M:** The RUI must act as an abstraction layer, hiding the internal logic of the ADAS subsystem (e.g., fusion algorithms, TTC calculation, decision logic) from the vehicle's components.
- **REQ-RUI-03 M:** The RUI must support the bidirectional flow of information necessary for emergency management, including:
    - (a) Acquiring vehicle state data (odometry, steering angle) and perception data (object tracking from sensors).
    - (b) Sending actuation commands (e.g., braking torque, steering torque) to the vehicle's controllers.

- **REQ-RUI-04 M** The interface must only expose the defined inputs and outputs (e.g., "request max braking torque").
- **REQ-RUI-05 M** The ADAS subsystem interface (RUI) must remain fully operational until State 4 (MANEUVER_COMPLETE) is reached or the vehicle comes to a complete stop in an emergency.

4.3.3. Relied Upon Physical Interface (RUPI) – Physical and Environmental Interface

The RUPI is the physical interface where energy or things are exchanged (e.g., physical observation via sensors) between the CSs.

- **REQ-RUPI-01 M:** Each CS must be equipped with apparatus (Sensors/Actuators) that allow it to observe and act upon the entourage (common physical environment).
- **REQ-RUPI-02 M:** The vehicle's sensors (radar, lidar, cameras) must operate as the RUPI, providing time-stamped observations of the environment's physical state.
- **REQ-RUPI-03 M:** Requirements for precision (value/time uncertainties) and sample rates of the sensors (RUPI) must be specified to enable an accurate description of the stigmergic channels.

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5. Traceability matrix

Requirements from upper layer document (identify precisely the document and its version)	**This SRS**
[Req-1] Assume a road, with 2 undivided lanes. Assume vehicles A and B on the same lane. A is in front, B is behind A starting from a certain initial distance	**REQ-CS-B-03 M** **REQ-RUPI-01 M**
[Req-2] B is proceeding at a higher speed than A.	**REQ-CS-B-03 M**
[Req-3] When B is sufficiently close to A, it performs an overtake.	**REQ-CS-B-03 M** **REQ-CS-GEN-01 M** **REQ-CS-GEN-02 M** **REQ-CS-B-01 M**

	REQ-CS-B-03 M REQ-RUI-01 M REQ-RUI-02 M REQ-RUI-03 M REQ-RUI-04 M REQ-RUI-05 M REQ-RUPI-02 M REQ-RUPI-03 M
[Req-4]If a vehicle C is coming in the opposite direction, B and C must operate to avoid collision	REQ-SoS-01 M REQ-SoS-02 M REQ-SoS-03 M REQ-CS-GEN-01 M REQ-CS-GEN-02 M REQ-CS-B-02 M REQ-CS-C-01 M REQ-CS-C-02 M REQ-CS-C-03 M REQ-RUI-02 M REQ-RUI-03 M REQ-RUI-04 M REQ-RUPI-02 M REQ-RUPI-03 M